

EE216 Homework3

Introduction

In this lab you are required to:

- Design a linear-phase 15-stage FIR filter with AXI-Stream interface.
- Generate bitstream and test on **ZCU104**.

If you don't know what is a FIR filter, you can reference

- <https://www.cnblogs.com/alifpga/p/7902759.html>
- https://blog.csdn.net/Reborn_Lee/article/details/82891161

Declaration of the top-level

Please use the entity declaration as below:

```
module myFIR(  
    input clk,  
    input rst_n,  
    input S_AXIS_TVALID,  
    output S_AXIS_TREADY,  
    input S_AXIS_TLAST,  
    input [7:0] S_AXIS_TDATA,  
    input M_AXIS_TREADY,  
    output M_AXIS_TVALID,  
    output M_AXIS_TLAST,  
    output [31:0] M_AXIS_TDATA  
);  
parameter signed [4:0] h0 = 5'd1;  
parameter signed [4:0] h1 = -5'd2;  
parameter signed [4:0] h2 = 5'd3;  
parameter signed [4:0] h3 = -5'd4;  
parameter signed [4:0] h4 = 5'd5;  
parameter signed [4:0] h5 = -5'd6;  
parameter signed [4:0] h6 = 5'd7;  
parameter signed [4:0] h7 = -5'd8;
```

You should use the given parameter for your FIR filter.

MultiCycle Adder

The following code shows the traditional ripple carry adder

```
module FullAdder(  
    input Ci, input A, input B,  
    output S, output Co  
);  
assign {Co, S} = A + B + Ci;  
endmodule  
  
module HalfAdder(  
    input A, input B,  
    output S, output Co
```

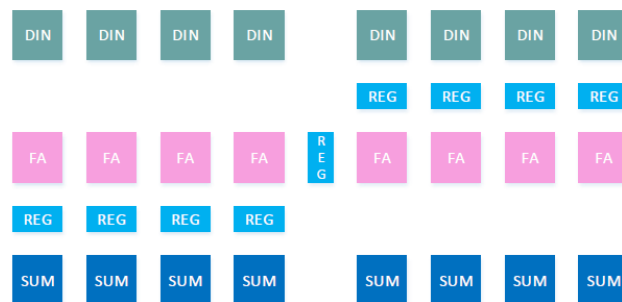
```

    input A,input B,
    output S,output Co
);
assign {Co,S}=A+B;
endmodule

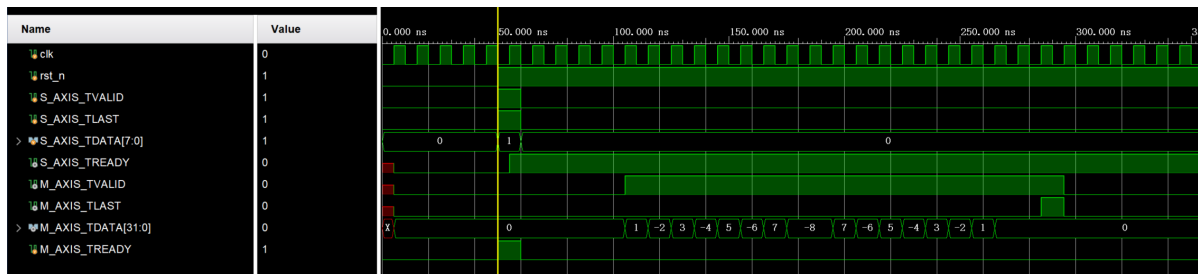
module FullAdder4(
    input Ci,
    input [3:0] A,
    input [3:0] B,
    output [3:0] S,
    output Co
);
wire [3:0] P;
wire [3:0] C;
wire BP;
assign P[0]=A[0] ^ B[0];
assign P[1]=A[1] ^ B[1];
assign P[2]=A[2] ^ B[2];
assign P[3]=A[3] ^ B[3];
assign BP=P[0]&P[1]&P[2]&P[3];
FullAdder FA0(Ci,A[0],B[0],S[0],C[0]);
FullAdder FA1(C[0],A[1],B[1],S[1],C[1]);
FullAdder FA2(C[1],A[2],B[2],S[2],C[2]);
FullAdder FA3(C[2],A[3],B[3],S[3],C[3]);
assign Co = BP ? Ci:C[3];
endmodule

```

You should add registers between the datapath of traditional ripple carry adder and reduce the critical path delay. This time, you should combine your multi-cycle adder with your FIR filter, the critical path delay should not exceed **one** addition and **one** multiplication.



The waveform should be like



Questions

- Why the output is delayed by 5 cycles after the first input valid ?
- What is the minimum data width of the output ?

Requirements

1. Implement the FIR filter using multi-cycle adder and AXI-Stream interface.
2. Write the testbench to test your code.
3. Generate bit stream and test on the board.

Build the bitstream

1. Extract the `hw2.zip`.
2. Put your source code (`myFIR.v`, `MultiCycleAdder.v`) in the `hw2/src` folder.
3. Open CMD, change the directory to `hw2` using `cd`.
4. Setup the environment

```
c:\xilinx\vitis\2020.2\settings64.bat
```

5. Run script to generate block design

```
vivado -mode batch -source ./scripts/block_design.tcl -notrace
```

6. Run scripts to generate bitstream

```
vivado -mode batch -source ./scripts/bitstream.tcl -notrace
```

7. When finish, find the `fir15.bit` and `fir15.hwh` in `hw2` folder.

Test on board

Connect to the board

You can use any SSH tools to connect to the board.

- IP address: 10.19.138.63
- Port: 9022
- Username: root
- Password: ee2162025

For public use, please create your own folder and do not save important files on the board, the board can be cleaned at any time.

Run the tests

1. Upload the `fir.bit`, `fir.hwh`, and `fir.py` to the board.
2. Run the test

```
python3 fir.py
```

Submission: Deadline on 2025-11-13 23:59 PM

You should submit:

1. **myFIR.v**: Verilog code of your FIR filter.
2. **MultiCycleAdder.v**: Code of multi-cycle adder.
3. **myFIR_tb.v**: Testbench of the top design.
4. **myFIR.bit**: Bitstream file
5. **myFIR.hwh**: Hardware handoff
6. **report.pdf**: A brief report in **PDF** format which including:
 - Screenshot of your waveform.
 - Answers to the questions.

Please compress all the files into a ZIP file with name **HW2_YourName_StudentNumber.zip**.

The file structure should be like:

```
HW2_张三_7654321/  
├─ myFIR.v  
├─ MultiCycleAdder.v  
├─ myFIR_tb.v  
├─ myFIR.bit  
├─ myFIR.hwh  
└─ report.pdf
```

Overdue Policy

- Submit on time, get all scores.
- Submit within 24 hours of the deadline, get half of the score.
- Submit more than 24 hours after the deadline, get 0 score.